

FRC ROBOT PROJECT - FINAL FROZEN BACKBONE ARCHITECTURE

FRC Java Coding Lab 7.0 - Permanent Inheritance Development Contract

Driver -> Xbox Controller -> controls -> commands -> subsystems -> io -> Hardware

telemetry observes beside the pipeline; it never controls the pipeline

PACKAGE TREE	RESPONSIBILITY MAP	FINAL FROZEN RULES
<div>frc.robot</div> <div><div><div>- Main / Robot / RobotContainer / Constants</div><div>- commands/ controls/ subsystems/</div><div>- io/</div><div><div>`- <mechanism>/</div><div><div>  - <Mechanism>IO</div><div>  - <Mechanism>IOReal</div><div>  `-<Mechanism>IOSim</div></div></div><div>- telemetry/</div><div><div> - RobotTelemetry</div><div><div>`- <mechanism>/</div><div><div>  - <Mechanism>TelemetryFacade</div><div>  `- focused telemetry modules</div></div></div><div>`- util/</div></div></div><div><div>deploy/pathplanner/</div><div><div>- paths/</div><div>`- autos/</div></div></div></div>	controls = Driver intent commands = Robot actions / coordination subsystems = Mechanism API and state io = Hardware contract + vendor APIs telemetry = Observation and diagnostics util = Generic shared helpers only SOURCE OF TRUTH • Hardware values originate in IO. • Mechanism state originates in subsystems/estimators. • Telemetry only publishes snapshots. • Each IO owns a mechanism-specific Inputs snapshot.	• RobotContainer: create, select, inject, configure. • Commands receive the smallest dependency needed. • Subsystems depend on IO interfaces only. • IO never publishes NetworkTables. • Telemetry never reads vendor devices or controls behavior. • Facade hides internal telemetry modules. • Constants.java is default; constants/ requires formal review. • commands/auto coordinates; PathPlanner files live in deploy/. • New mechanisms follow responsibility patterns, not file-by-file copying. • Normal lessons extend the backbone; they do not redesign it.

APPROVED FLOWS

CONTROL	OBSERVATION	FORBIDDEN
Controller -> controls -> command -> subsystem -> IO -> hardware	IO Inputs / mechanism state -> telemetry facade -> typed NT4	Telemetry -> control IO -> NT RobotContainer -> periodic logic

STATUS: FINAL FROZEN AND LOCKED - NO FURTHER BACKBONE CHANGES